

A Manikho Rajina

manikhorajina751@gmail.com — +91 8837282410 — linkedin.com/in/manikho — IIT Bombay, Mumbai, India

EDUCATION

- **IIT Bombay** Mumbai, India
PhD; CPI: 8.79/10
July 2023 - Present
Courses: Petroleum Exploration, Petrophysics, Seismology and Earth Structure, Computational Geophysics, Numerical Analysis
- **IIT Bombay** Mumbai, India
Master of Science; CPI: 7.93/10
July 2021 - June 2023
Courses: Exploration Seismology, Geophysical Inverse Theory, Computer Programming, Mathematical Physics, Remote Sensing and GIS, Numerical Analysis

SKILLS

- Languages: Python, Matlab, C, Bash
- Tools: SAC, GSAC, GMT, SeiSee, ArcGIS, Seisan, Google Earth Pro
- Platforms: Linux (HPC clusters), Web, Windows, Mac
- Soft Skills: Leadership, Mentoring, Team Collaboration, Public Speaking, Time Management

INTERNSHIP & TRAINING

- **Petroleum Engineers Association** Remote
Internship (Part-time)
Jul 2022 - Aug 2022
 - **Reservoir Permeability:** Developed code for estimating reservoir permeability from pressure-time data of drawdown well tests.
 - **Data Interpretation:** Gained hands-on experience in well log data interpretation using real field data.
 - **Production Engineering:** Gained exposure to production systems, well completion, and testing operations.
- **IIT Bombay** Rajasthan & Mumbai
Training (Part-time)
Jul 2023 - Jul 2025
 - **Instrument Deployment:** Deployed multiple seismographs and Raspberry Shake 3D across the Aravalli region and Palghar for passive seismic monitoring.
 - **Seismic Survey:** Acquired 2D seismic refraction data and designed surveys using a 24-channel geophone array.
 - **Electrical Survey:** Conducted DC resistivity surveys (sounding & profiling) and applied forward modelling and inversion to map subsurface structures.
 - **Gravity & Magnetic Survey:** Acquired gravity and magnetic data using CG-5 Autograv gravimeter and Proton Precession Magnetometer, integrated with DGPS for elevation measurement.

PROJECTS

- **PhD - Shallow Crustal Structure of the Indian Continent Using Transdimensional Bayesian Inversion** (Ongoing)
 - Built automated Python workflows for receiver function analysis and P-wave polarization, improving subsurface imaging in complex geological settings.
 - Developed and implemented a transdimensional Bayesian joint Markov Chain Monte Carlo (MCMC) inversion approach, successfully delineating sub-basalt sedimentary structures.
- **MSc - Ambient noise seismic interferometry of continuously recorded seismic data from the Cachar fold belt Assam** (Mar '23)
 - Imaged hidden 3D subsurface structures in the Cachar fold belt using ambient noise tomography.
 - Enhanced exploration planning by guiding active seismic surveys with passive imaging results.

POSITIONS OF RESPONSIBILITY

- Interview Coordinator: Coordinated a 250⁺ member team to streamline interviews for 1800⁺ students, supporting recruitment processes for 20⁺ firms and managing candidate queries (Dec '22).
- Vice-President IGU: Mentored 100⁺ students in geoscience careers and organized workshops, seminars, and industry networking events (May '23).
- Teaching Assistant: Assisted in seismological data acquisition, processing, and interpretation (Present).
- MSc Supervision: Supervised two master's students on seismological data processing and inverse modelling (Mar '25).

ACHIEVEMENTS

- Qualified GATE 2024 and 2023 (Geology & Geophysics) in consecutive years.
- Qualified IIT-JAM 2021 (Physics) among 15,000⁺ candidates.
- Consecutive Chess Champion representing IIT Bombay in AICL (All India Chess League 3.0 & 4.0), competing with 1,000⁺ players from 30⁺ universities nationwide.
- Secured 1st place in District Quiz Competition organized by All India Radio, Imphal.
- Won 3rd place in Quiz Competition organized by Guwahati University on National Science Day.
- Secured letter marks in Science, Mathematics, and regional language Mao in Matriculation.

CONFERENCE PRESENTATIONS

- Rajina, A. M. and Maurya, S: Sub-Basaltic Sediment Imaging with Teleseismic Earthquakes using a Transdimensional Bayesian Approach, EGU General Assembly 2026 (Poster Presentation, EGU26-21032).
- Rajina, A. M. and Maurya, S: Imaging sedimentary thickness and the Main Himalayan Thrust beneath the Hi-CLIMB array using Bayesian inference, 9th Annual Convention of "Advances in Earthquake Science", 2026 (Oral Presentation).
- Rajina, A. M., Maurya, S., and Mohan, G: Sediment and Basement Architecture in Northeast India and Bangladesh from Teleseismic P-wave coda using a Bayesian Monte Carlo approach, 16th General Assembly of the Asian Seismological Commission, 2026 (Poster Presentation).

CERTIFICATIONS

- Completed IWCF Level-1 and Well Operations Crew Resource Management (WOCRM).
- Completed a course on Machine Learning Specialization by DeepLearning.AI, instructed by Andrew Ng.
- Completed a course on Digital Signal Processing (DSP), with a strong focus on Fourier analysis, DFT, and DTFT.
- Completed Convolutional Neural Networks in Python, with applications in image recognition using Keras and TensorFlow.
- Completed GSITI e-training on SP and gravity methods, focusing on survey design, data acquisition, and interpretation.